

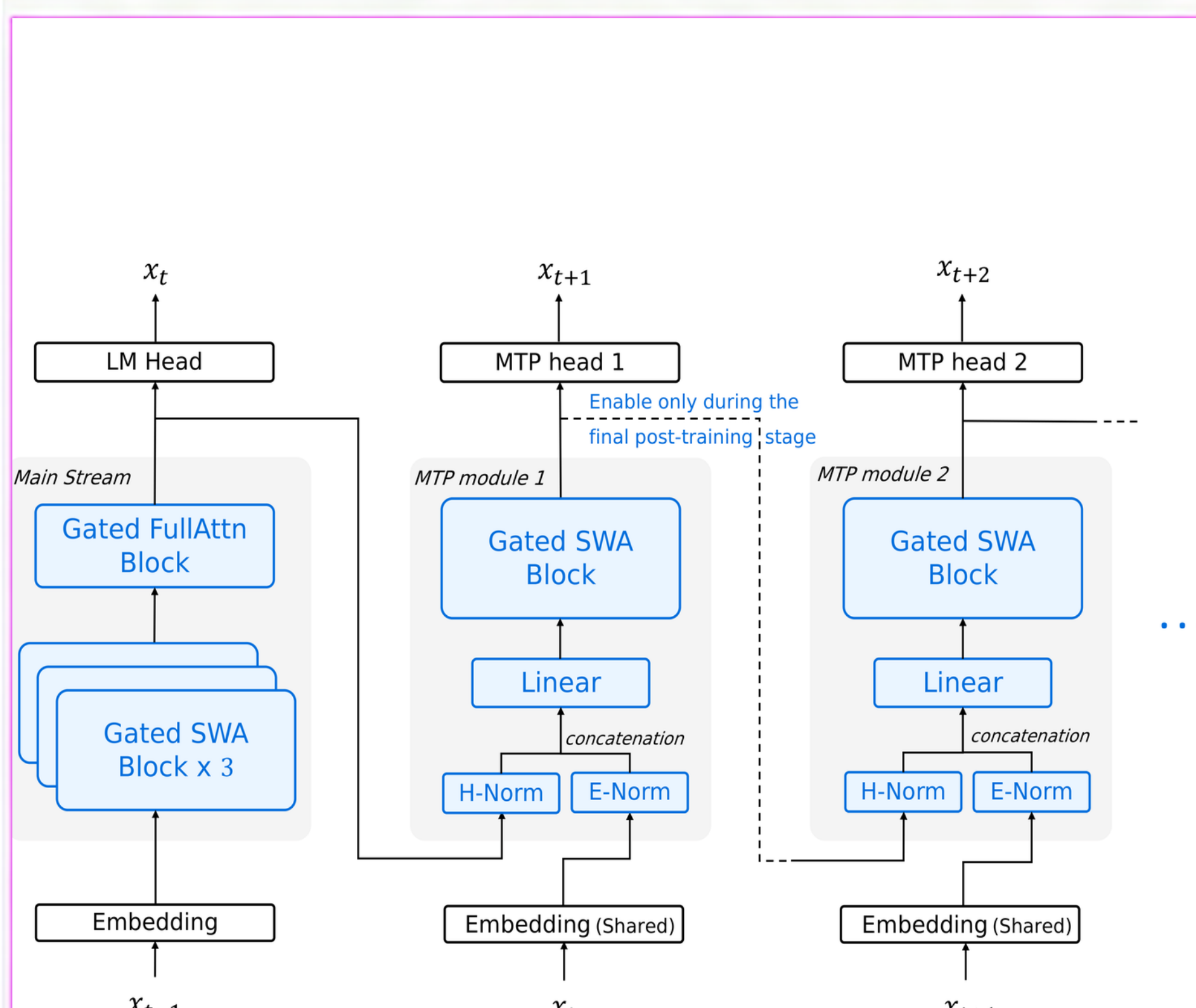


ONE-LINE TAKEAWAY: Step 3.5 Flash achieves frontier-level intelligence with only 11B active parameters (196B MoE), matching leading models like GPT-5.2 xHigh and Gemini 3.0 Pro on reasoning and agentic tasks.

1. PROBLEM

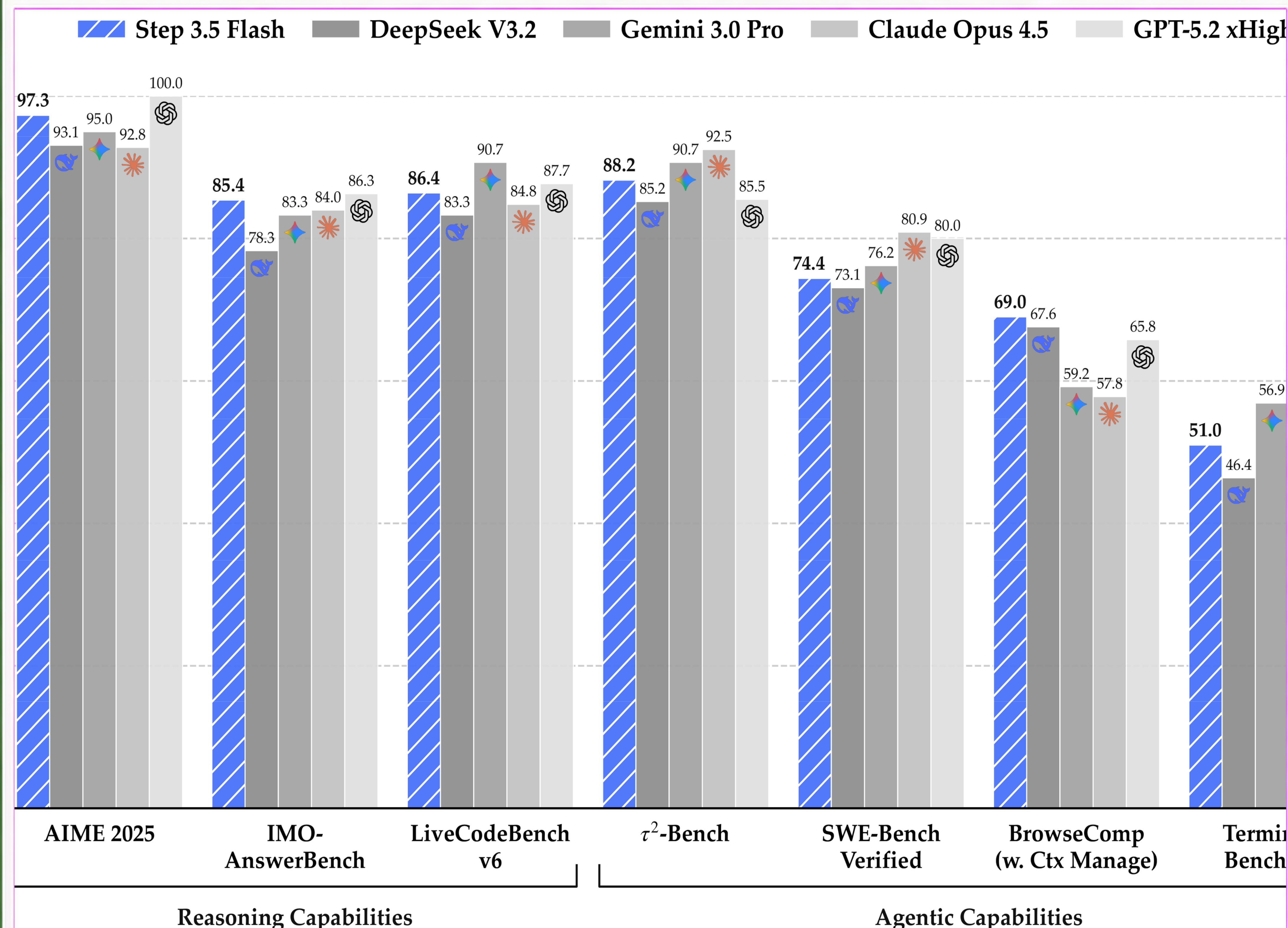
- Agentic systems require low latency for real-time interaction, but long-context inference is computationally expensive.
- Current post-training methods struggle with scalable reinforcement learning and efficient expert iteration in sparse Mixture-of-Experts models.

Figure 1: Step 3.5 Flash achieves frontier-level intelligence with only 11B active parameters (196B MoE)



2. METHOD / ARCHITECTURE

Figure 2: Illustration of Step 3.5 Flash. The model uses head-wise gated attention, hybrid attention, MTP-3, and MIS-PO for efficient and stable training.



3. RESULTS

- Uses a sparse Mixture-of-Experts (MoE) architecture with 196B total parameters and 11B active per token.
- Employs hybrid attention with 3:1 ratio of Sliding Window Attention (SWA) to Full Attention.
- Implements Multi-Token Prediction (MTP-3) for speculative decoding to reduce latency.
- Applies head-wise gated attention and GQA-8 for hardware-aligned efficiency.
- Uses Metropolis Independence Sampling-Filtered Policy Optimization (MIS-PO) for stable RL training.

BENCHMARK RESULTS (Accuracy)

IMO-AnswerBench	85.4%
LiveCodeBench-v6 (2024.08–2025.05)	86.4%
τ^2 -Bench	88.2%
BrowseComp (with Context Manage)	69.0%
Terminal-Bench 2.0	51.0%

KEY HIGHLIGHTS

 11B Active parameters during inference (out of 196B total)	 170 tokens/s Inference speed on Hopper GPUs during first week on OpenRouter	 85.4% Accuracy on IMO-AnswerBench
--	---	---

Figure 3: Per-step training loss of Step 3.5 Flash, plotted without smoothing or

